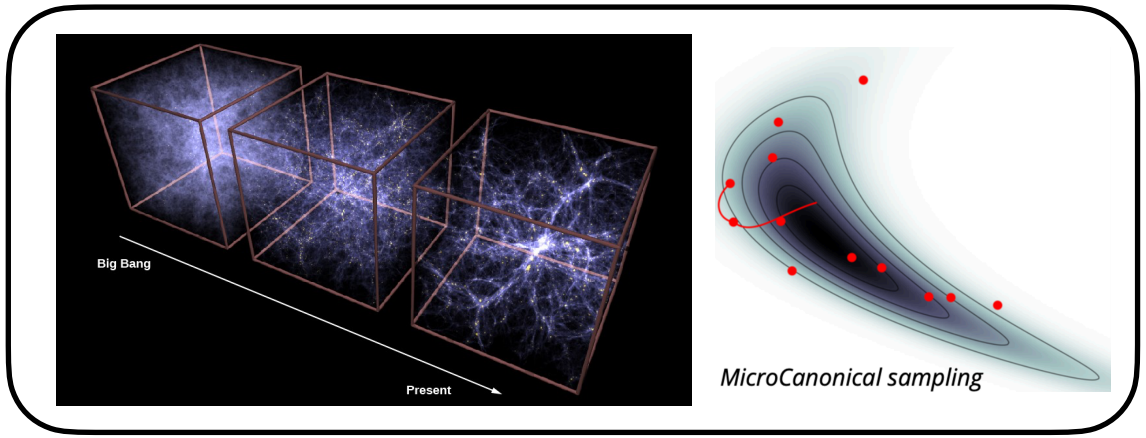
ADE · NOVEMBER 2026

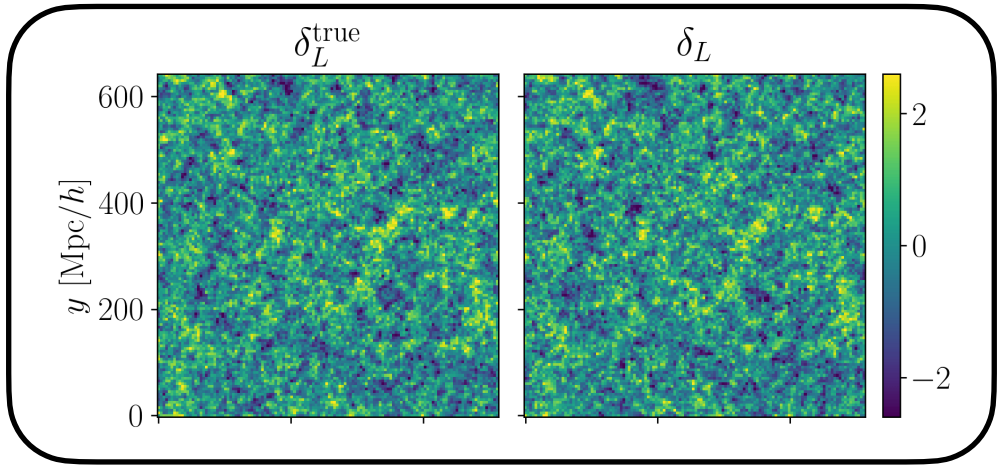
Galaxy clustering × CMB lensing at the field level

Jonathan Hawla

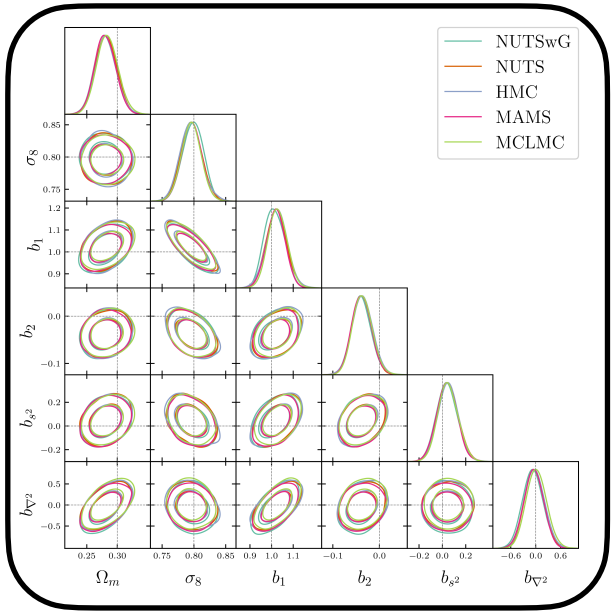
with Etienne Burtin, Arnaud de Mattia, Hugo Simon, Wassim Kabalan, Constantin Payerne

Field level inference





github.com/hsimonfroy/benchmark-field-level



Hugo Simon et al. 2025

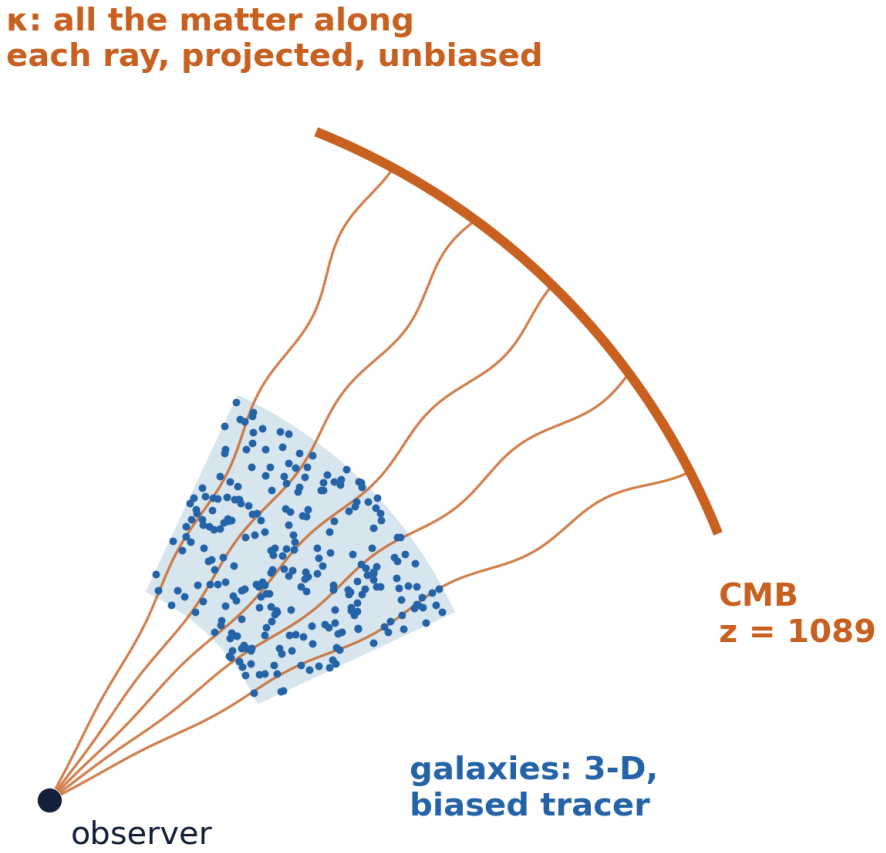
Why the field level, and why Galaxy clustering × CMB lensing?

Field level

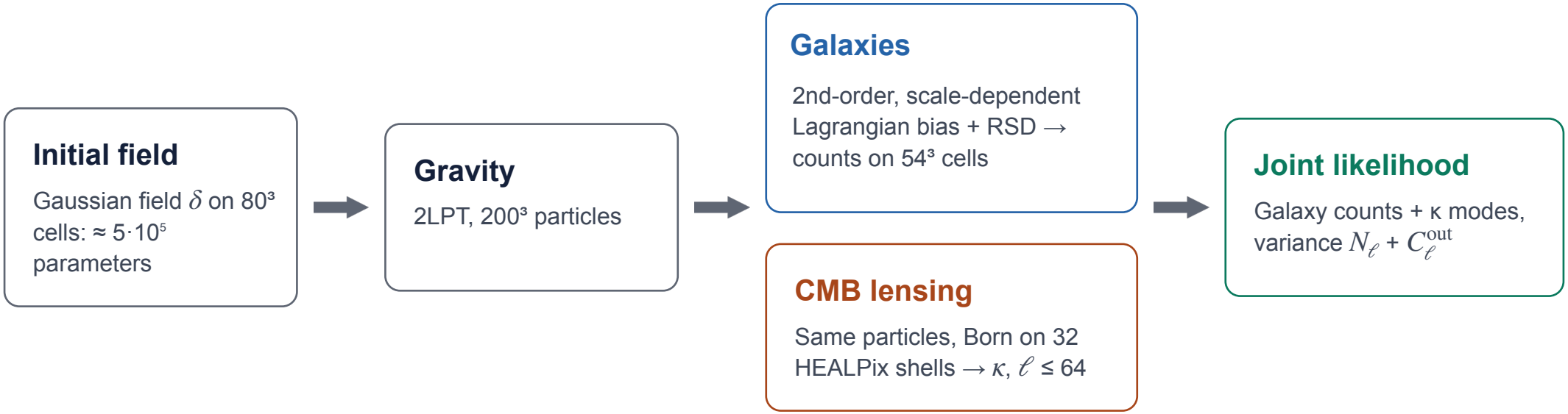
No summary statistic and **no mock covariance**.
Reconstruct the **initial field** in addition to the bias and cosmology.
Limitation: the volume and resolution of the 3D model.

CMB lensing κ

Probes all the matter, with no galaxy bias: it breaks the b_1-σ_8 degeneracy.

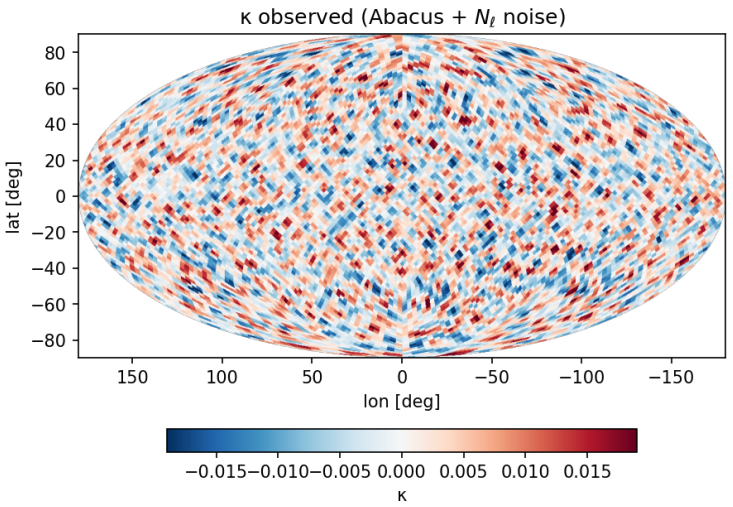


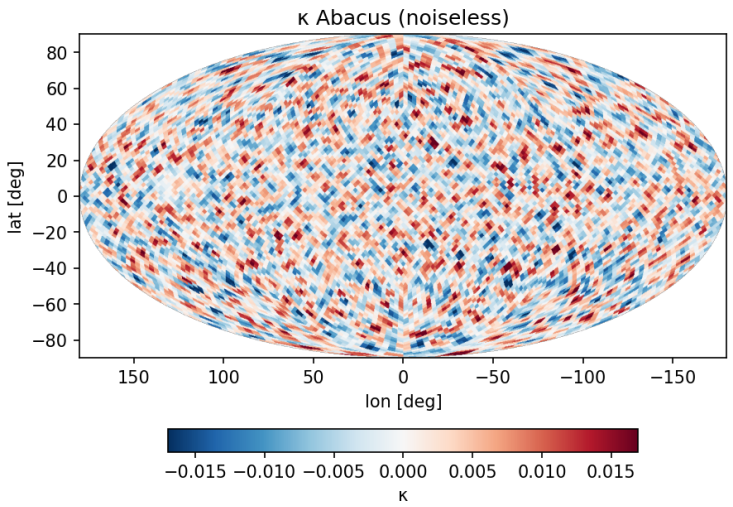
One differentiable forward model, two observables

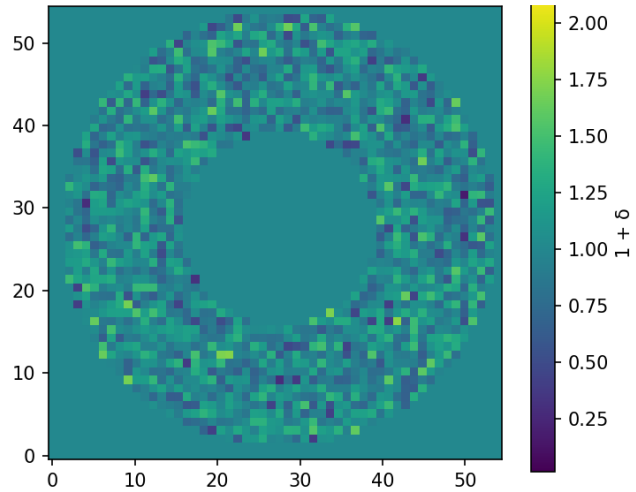


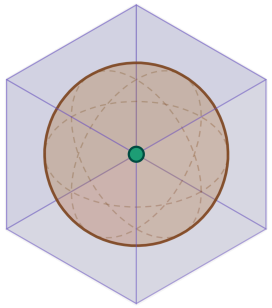
MCLMC, gradients by JAX · one chain run on an A100 (80 GB), 0.29 s per gradient, **total 5.4 h**

The test sky: AbacusSummit HUGE









Data: full-sky LRG light cone + full-sky CMB κ map simulation (Independent, High-resolution, N-body, HOD, Born Approximation)

Noise: Simons Observatory baseline N_ℓ, κ at ℓ 2–64

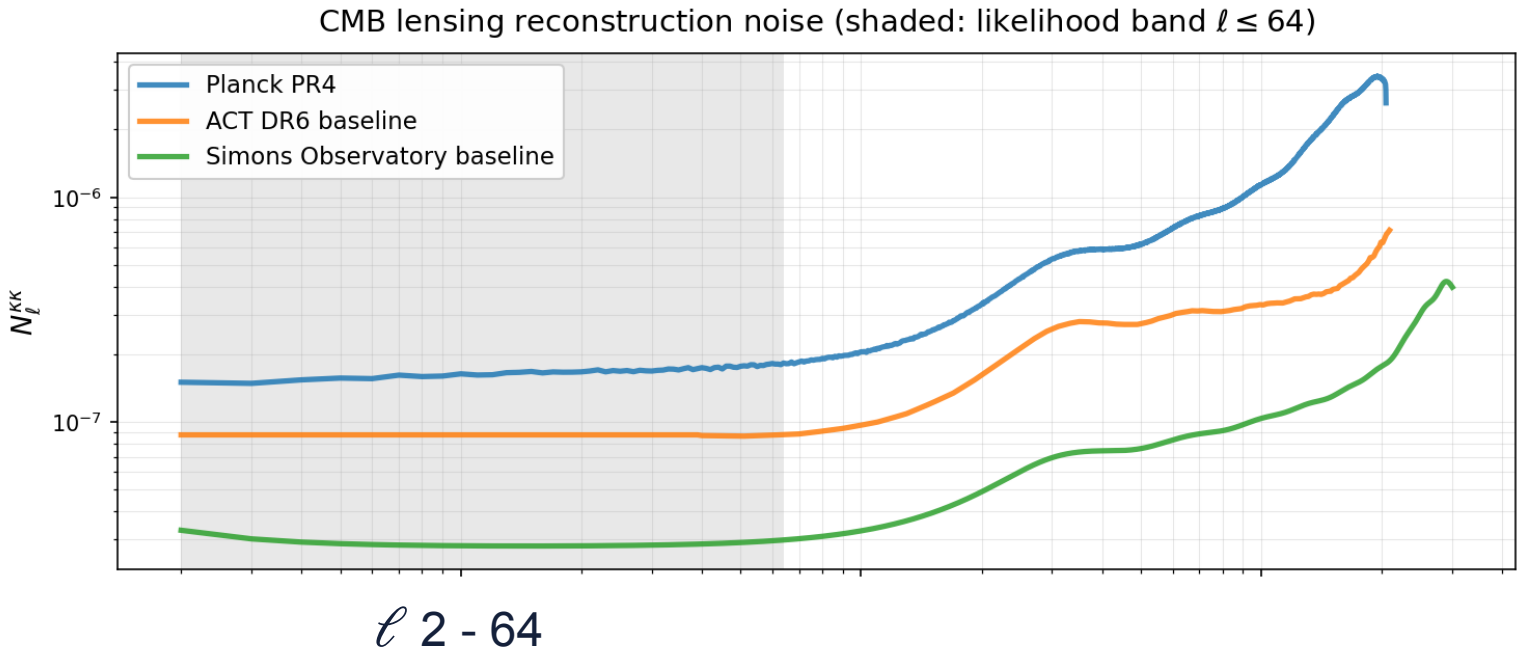
Goal: recover unbiased $\Omega_{\rm m}$, σ_8 and reconstruct the initial field

The joint likelihood

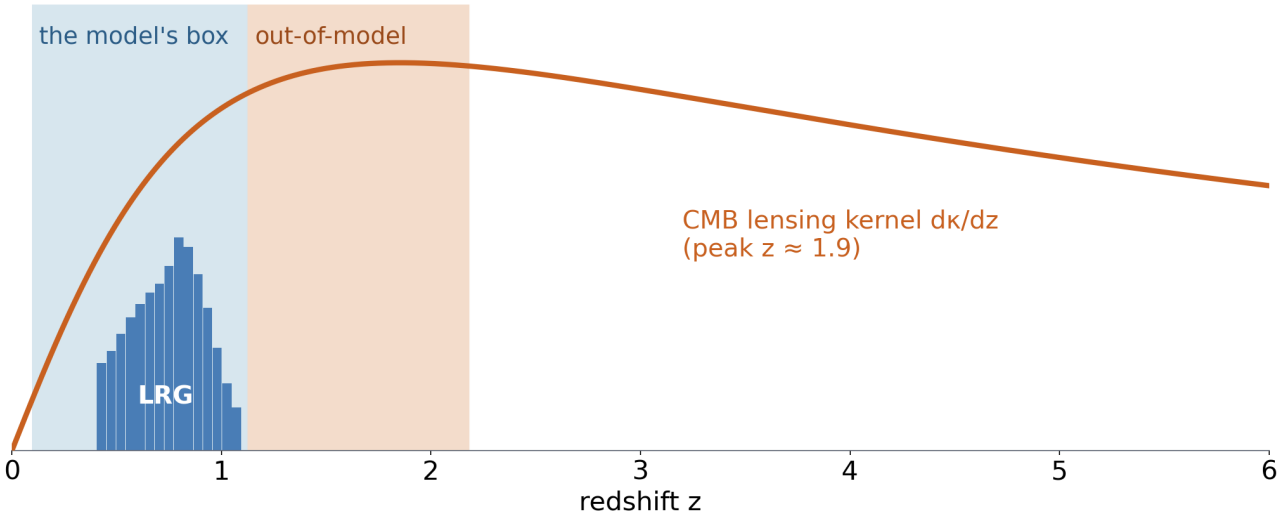
$$-\ln \mathcal{L} = \sum_{\text{cells}} \frac{[N_{\text{obs}} - \bar{n} S\,(1+\delta_g)]^2}{2\,\bar{n} S}$$

$$+ \sum_{\ell m} \frac{\left|a_{\ell m}^{\text{obs}} - a_{\ell m}^{\text{model}}\right|^2}{2\,C_\ell(\theta)} + \frac{1}{2}\ln C_\ell(\theta), \qquad C_\ell(\theta) = N_\ell + C_\ell^{\text{out}}(\theta)$$

Unmodeled power analytically marginalized as a Gaussian variance



The map holds κ beyond the model depth



PROBLEM

The model cannot produce matter beyond the box

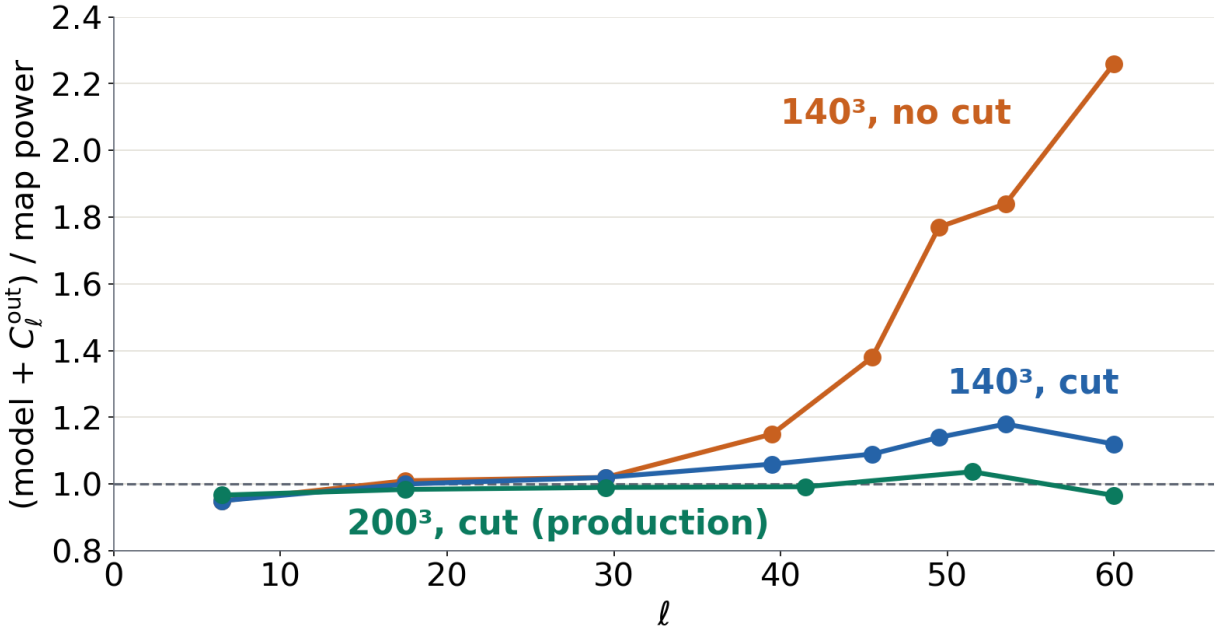
FIX

$$C_\ell^{\mathrm{high-z}}(\theta) = \int_{\chi_{\mathrm{end_box}}}^{\chi_{\mathrm{CMB}}} \frac{W_\kappa^2(\chi)}{\chi^2} \times P(k,\chi) d\chi$$

theoretical spectra of the missing power

$\rightarrow C_\ell^{\mathrm{out}}$

The map holds κ below the model resolution



PROBLEM

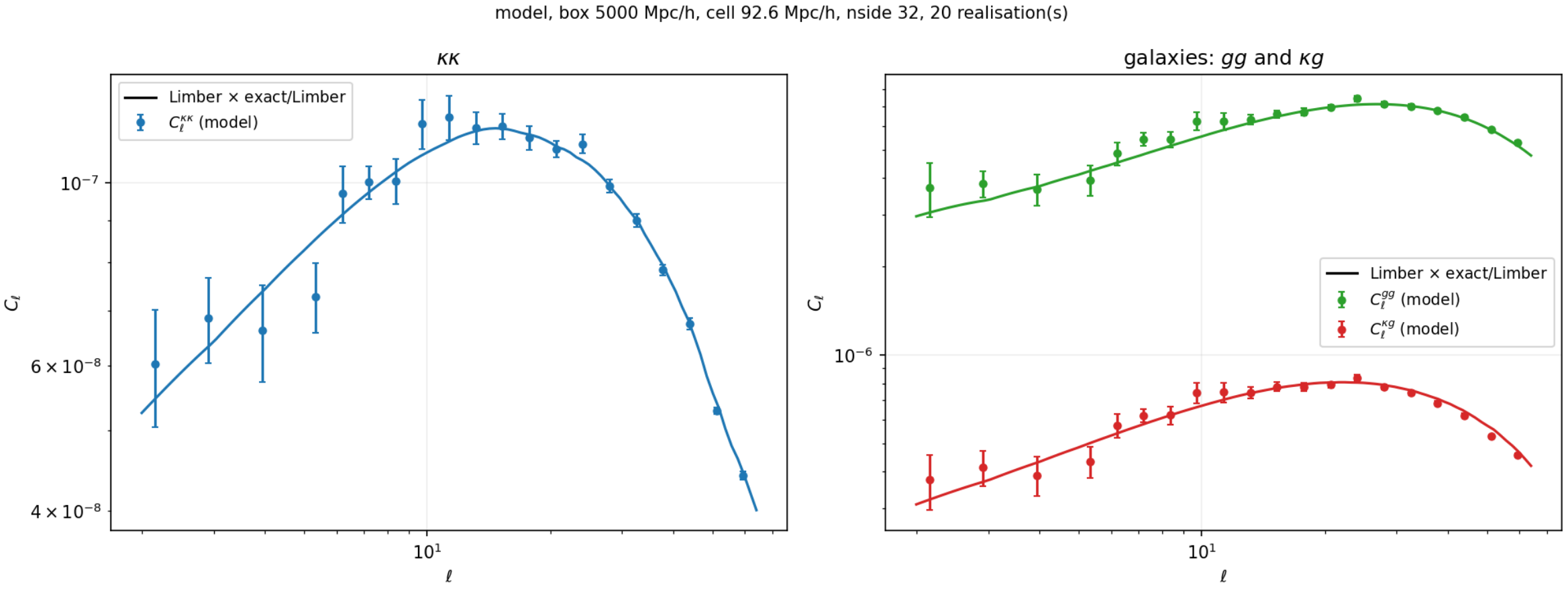
at high ℓ a **near shell** sees scales the field does not resolve

FIX (W. KABALAN)

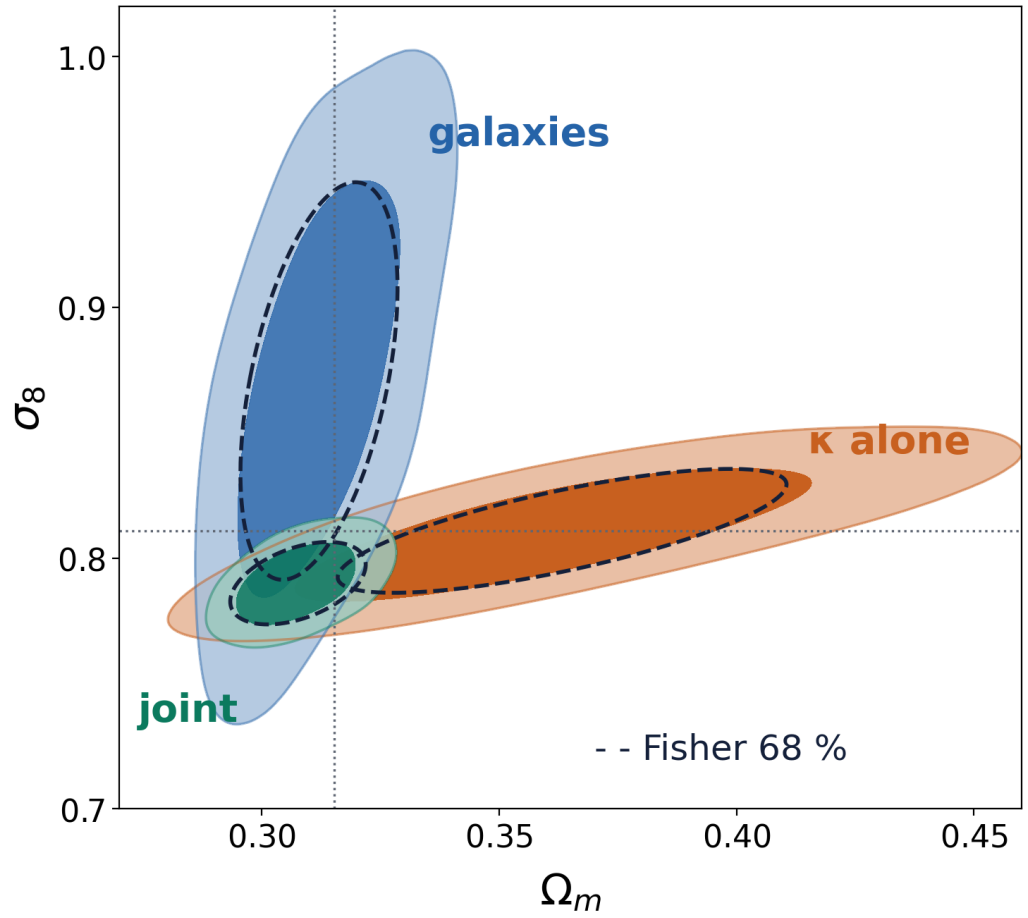
$$\ell_{\rm res} = k_N^{\rm init}\, r_{\rm eff}$$

each shell **low-passed** above $\ell_{\rm res}$, removed power → $C_\ell^{\rm out}$

The forward model against theory

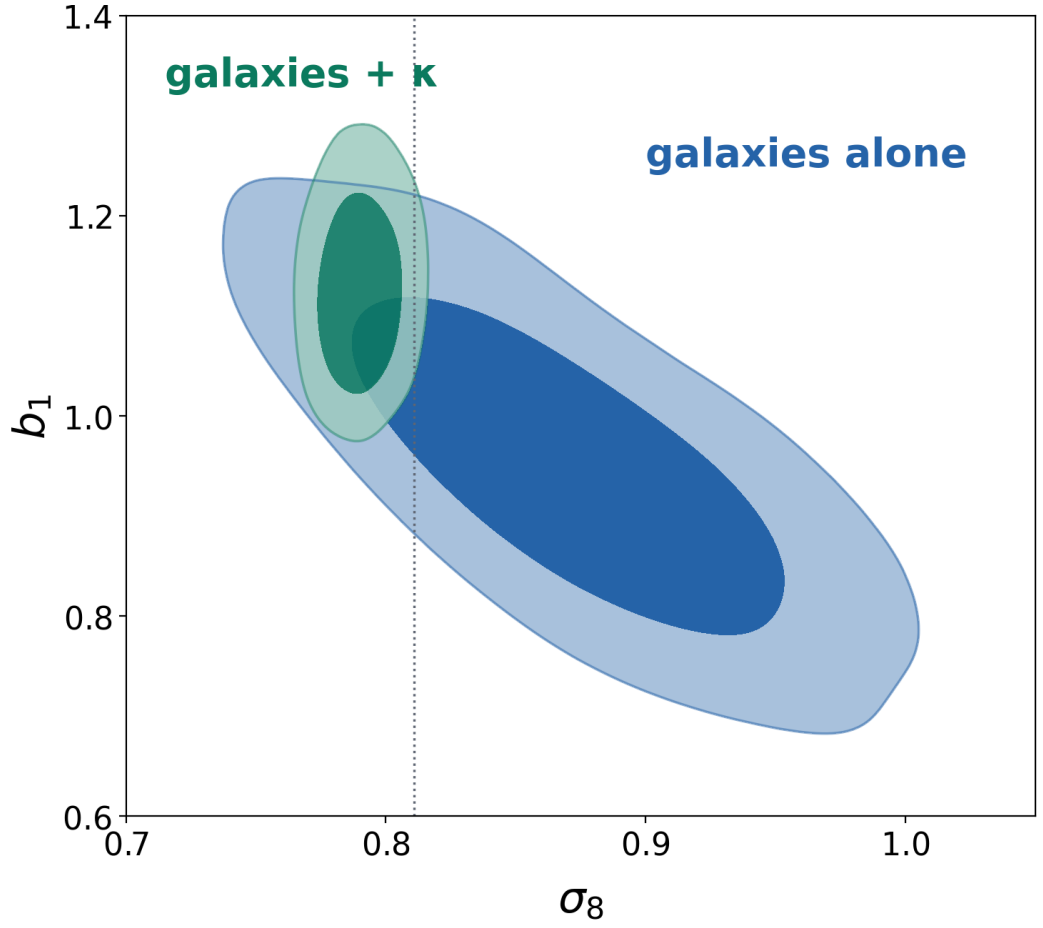


Result: galaxies, κ and joint on the Abacus sky

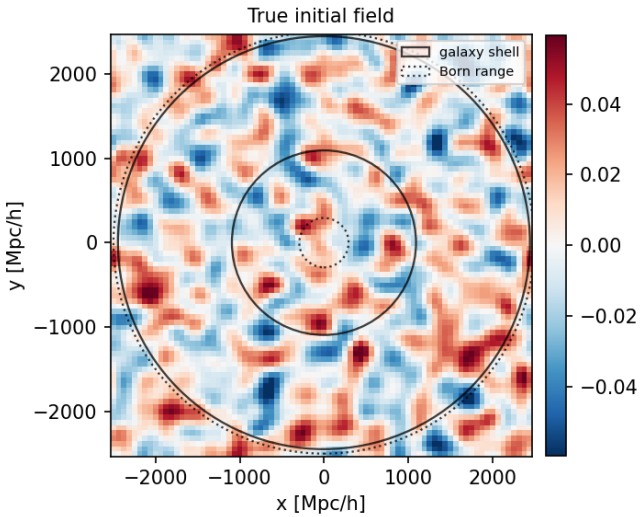


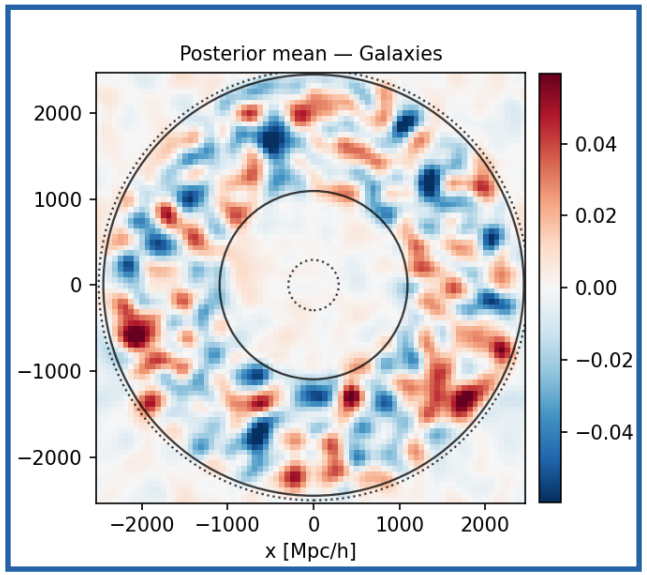
converged (split $\hat{R} \leq 1.006$); widths on the Fisher (dashed)

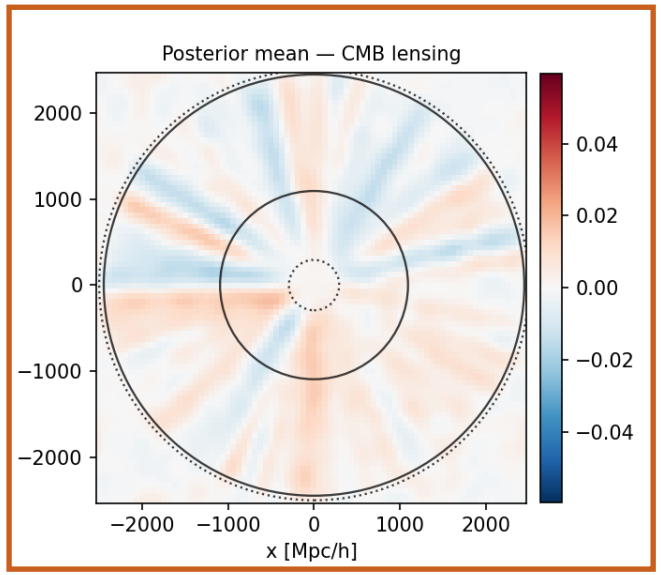
κ breaks the b_1–σ_8 degeneracy

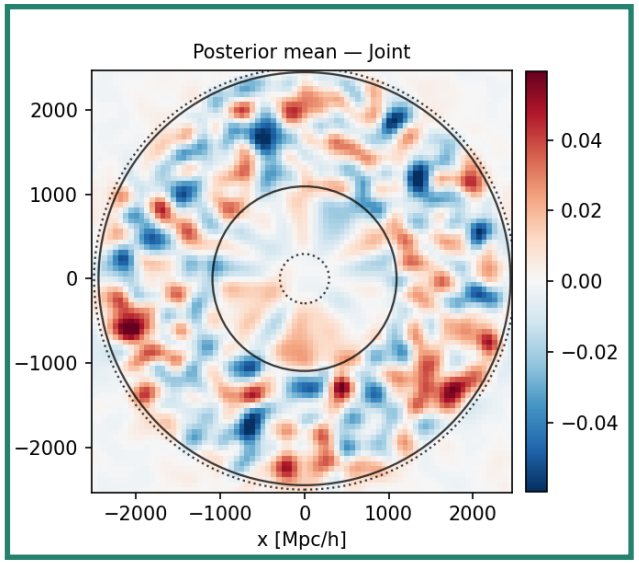


κ reconstructs the field in front of the galaxies

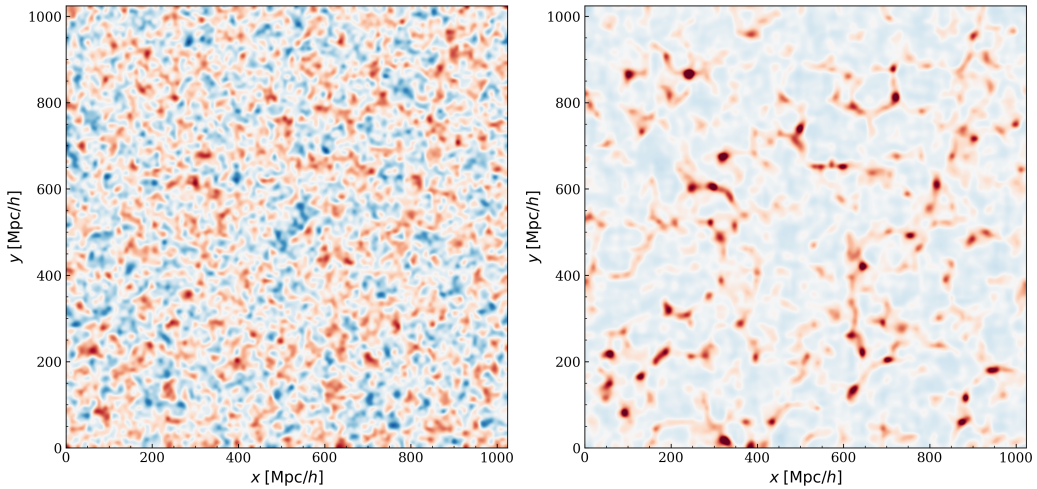








Next: toward the scales of standard analyses



At $k \approx 0.1$ the field is no longer Gaussian: with the same modes, the field level gives σ_8 **3.5× tighter** than P + B (Nguyen et al. 2024, $k_{\mathrm{max}} = 0.10$). Does it hold for galaxies × κ on Abacus?

Take-home

1. A joint CMB lensing × galaxy field-level inference runs on a realistic N-body sky, with $\Omega_{\rm m}$, σ_8 and six bias parameters free, and its posterior is consistent with the truth.

2. Each systematics we met had a physical cause, found by a targeted test, and a fix checked against the simulation's own initial conditions.

3. κ breaks b_1–σ_8 and reconstructs the matter in front of the galaxies.

Backup slides

Forward model: gravity and galaxies

Gravity: second-order Lagrangian perturbation theory

$$\mathbf{x} = \mathbf{q} + \boldsymbol{\Psi}(\mathbf{q}), \qquad \boldsymbol{\Psi} = -D_1 \nabla \phi^{(1)} + D_2 \nabla \phi^{(2)}$$

$\phi^{(1)}$ linear in δ (Zel'dovich), $\phi^{(2)}$ quadratic

Galaxies: a bias weight per particle, to second order

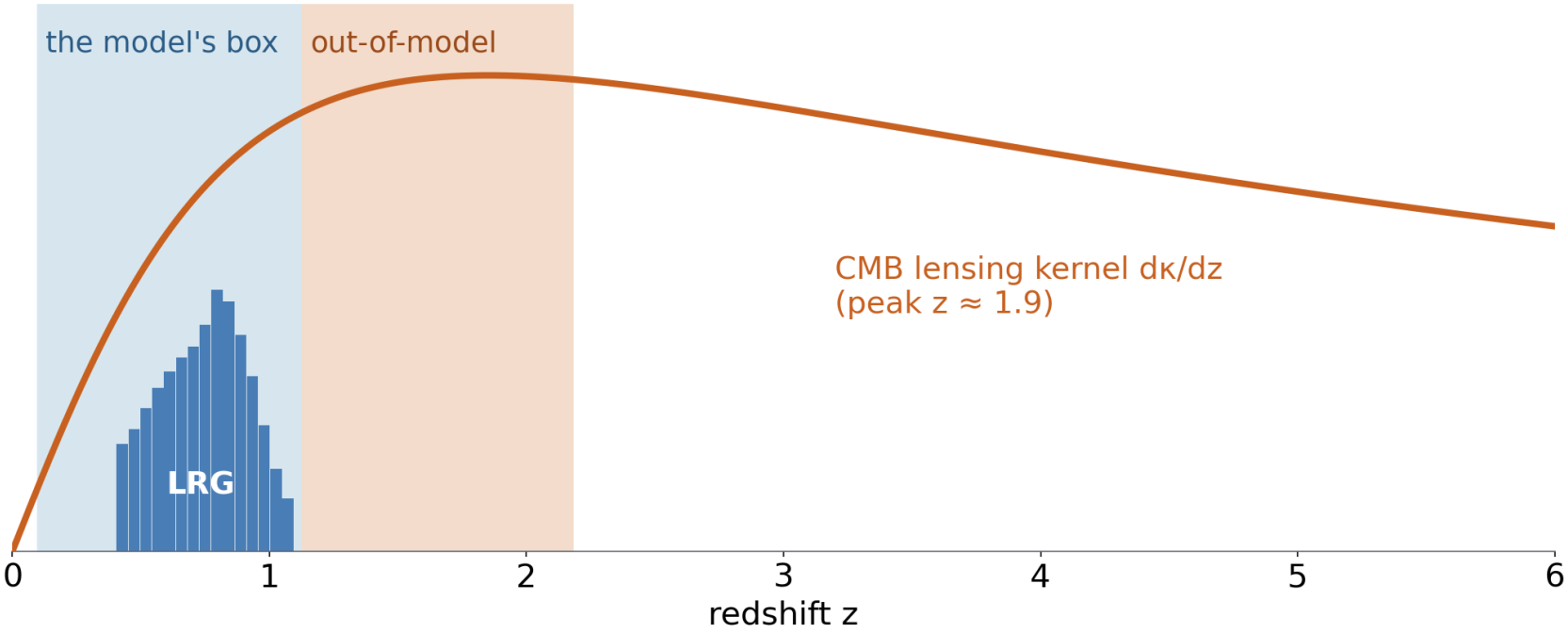
$$w(\mathbf{q}) = 1 + b_1\,\delta_L + \frac{b_2}{2}\left(\delta_L^2 - \langle \delta_L^2 \rangle\right) + b_{s^2}\left(s^2 - \langle s^2 \rangle\right) + b_{\nabla^2}\,\nabla^2 \delta_L$$

weighted particles moved to redshift space (RSD, Finger-of-God $b_{\nabla\parallel}$)

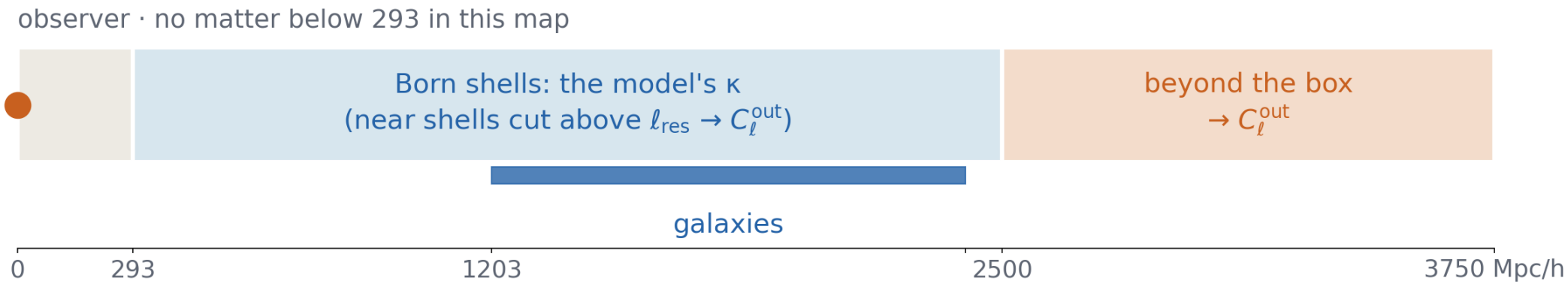
b_1 depends on the redshift, inversely proportional to $D(\chi)^\alpha$, with α a free parameter

Forward model: CMB lensing (Born)

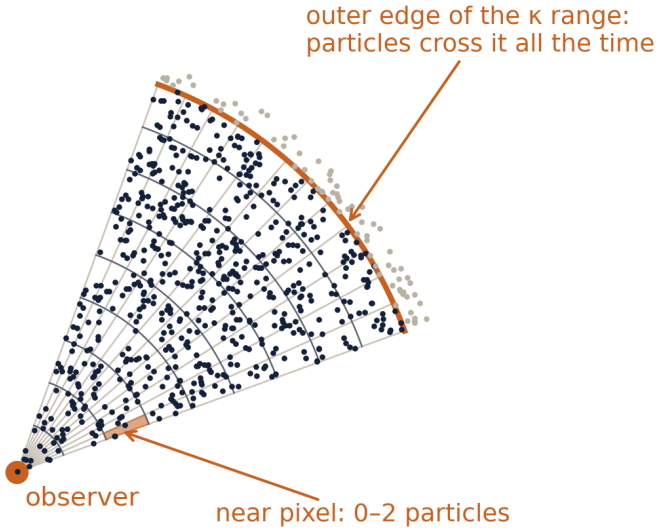
$$\kappa(\theta) = \int d\chi W_\kappa(\chi)\delta_m(\chi,\theta) \qquad W_\kappa(\chi) = \frac{3}{2}\Omega_{\rm m}\left(\frac{H_0}{c}\right)^2 \frac{\chi}{a}\frac{\chi_s-\chi}{\chi_s}$$

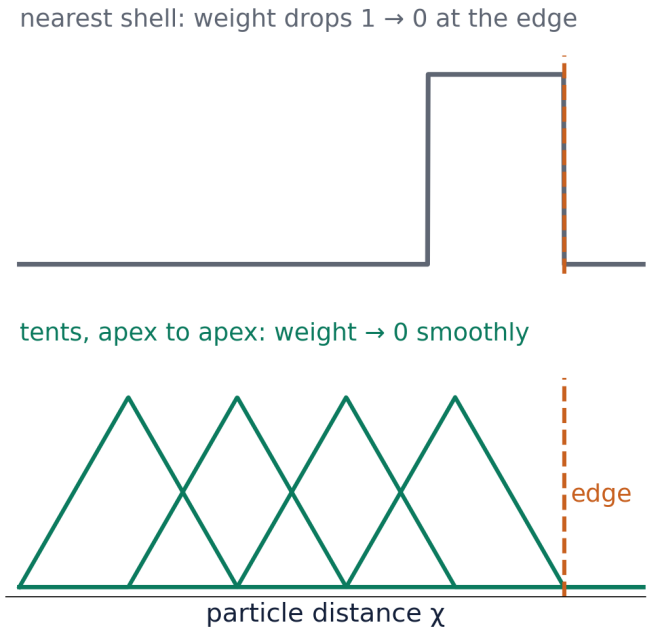


Out of model in summary



CMB lensing made the joint unsampleable





PROBLEM

κ divided the joint step size by ≈ 30

CAUSE

log p jumps as particles leave the κ range at its outer edge

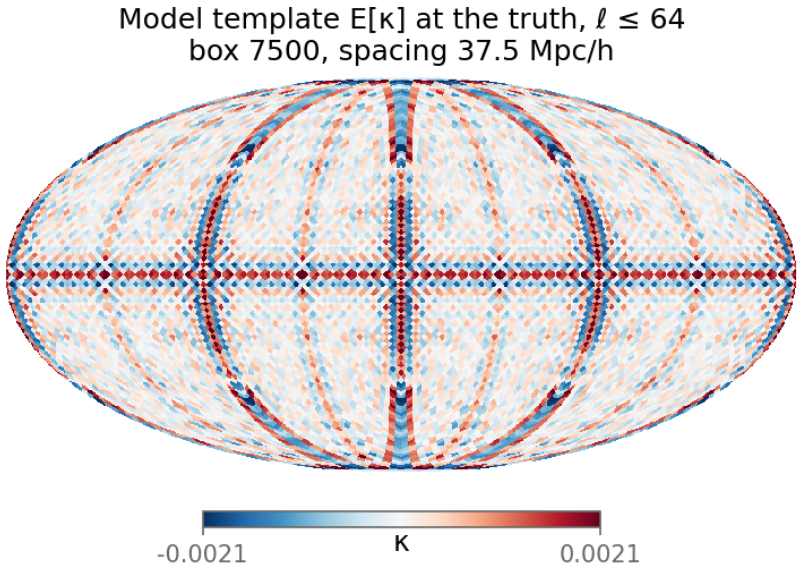
FIX

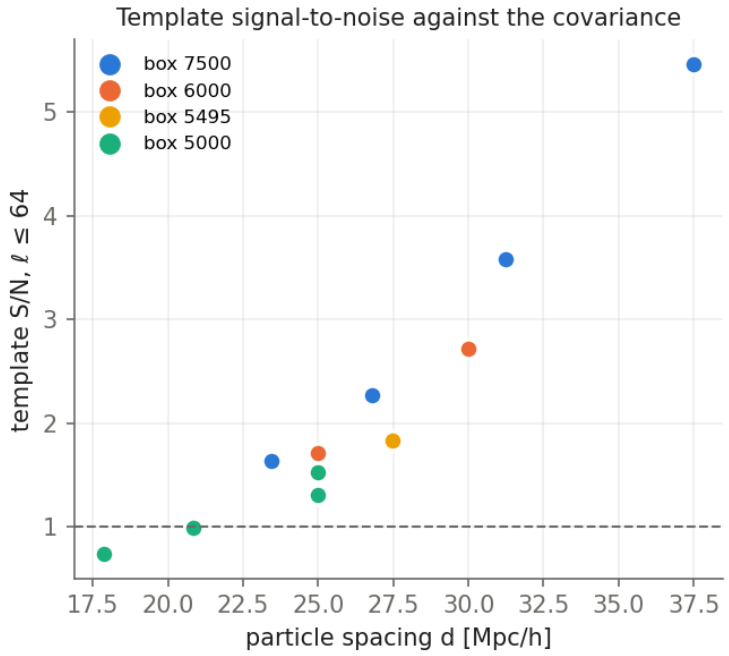
radial tents, apex to apex; near shells: the ℓ cut

CHECK

joint step 85.2, galaxies alone 85.9

The particle lattice leaves a pattern in κ





PROBLEM

κ alone: Ω_m 0.236 for a truth of 0.315

CAUSE

a fixed lattice pattern: 1 % of the power, read on every mode

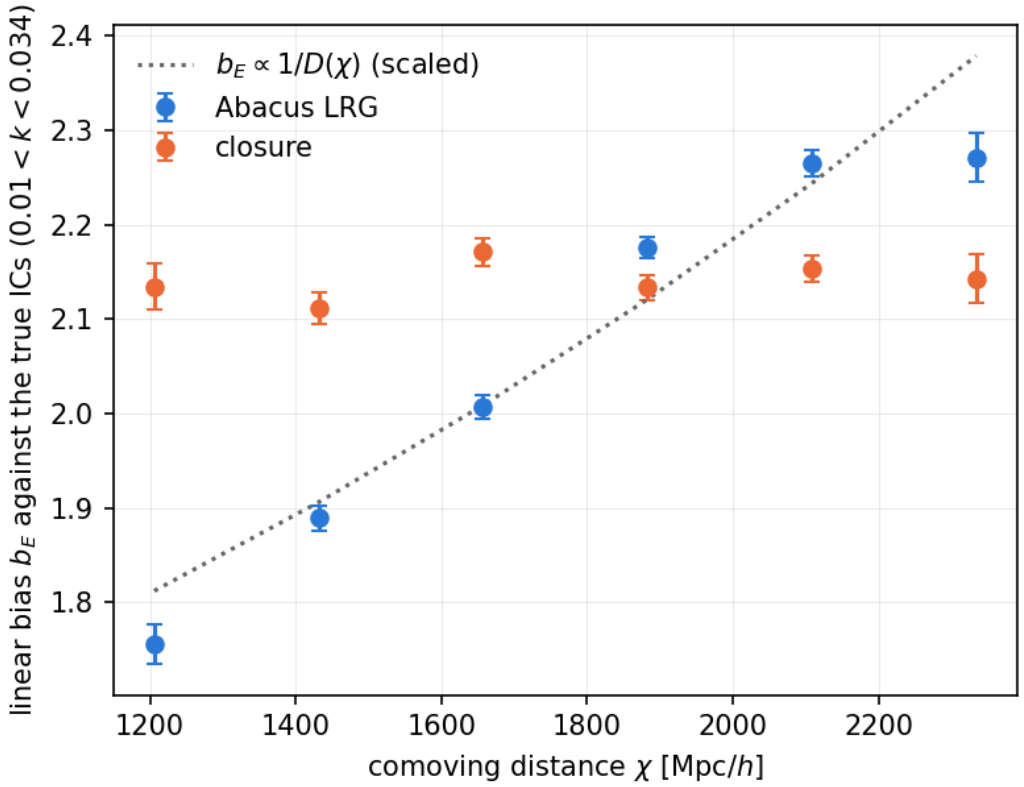
FIX

particle spacing 25 Mpc/h: 200³ in a 5000 Mpc/h box

CHECK

S/N 5.5 → 1.3

The LRG bias evolves with redshift



PROBLEM

galaxies alone: Ω_m 0.246, σ_{8_8} 0.603

CAUSE

bias grows as $1/D$; a single b_1 bends Ω_m instead

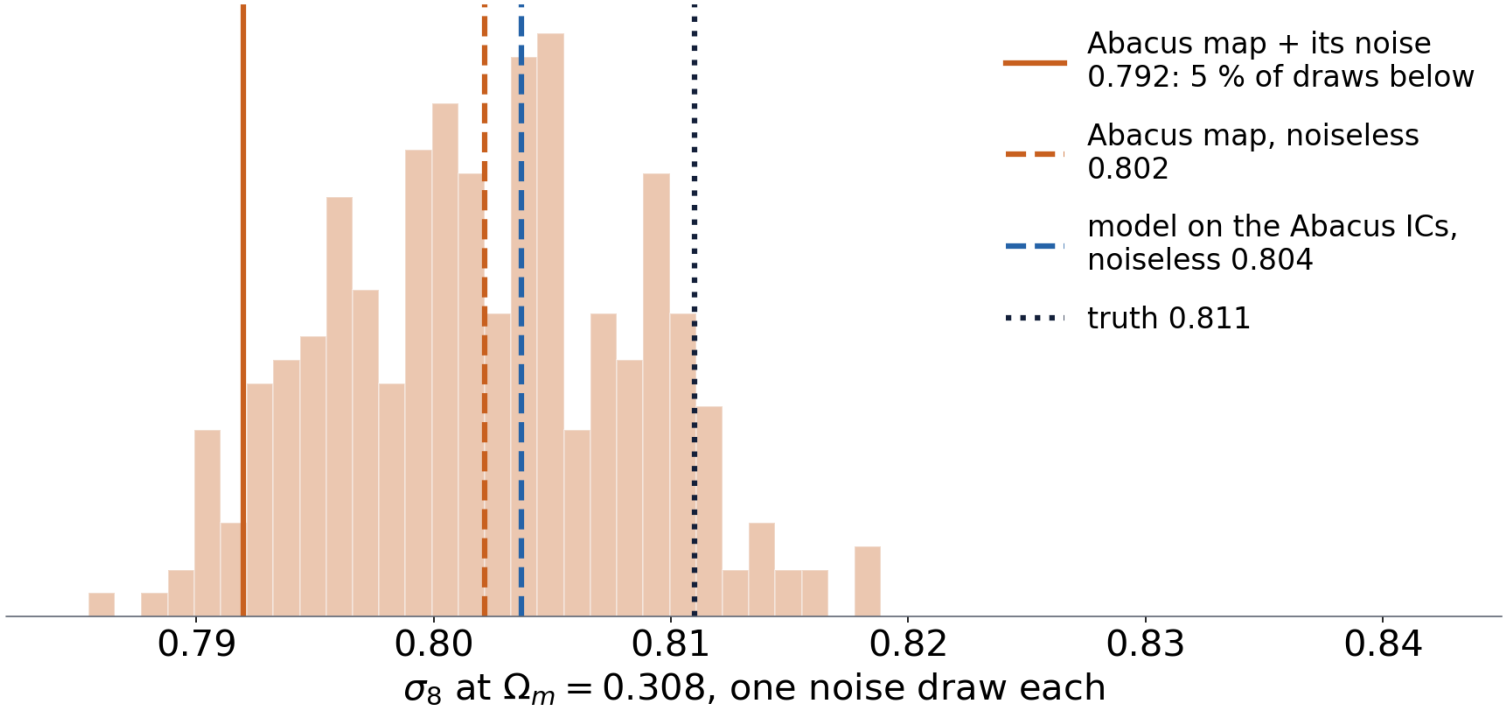
FIX

free bias evolution α, and 200³ particles

CHECK

Ω_m −0.3σ, σ_8 +1.1σ, α = 1.03 ± 0.04

The joint σ_8 is 2σ low: model error?



PROBLEM

joint σ_8 = 0.790 ± 0.011, -2σ

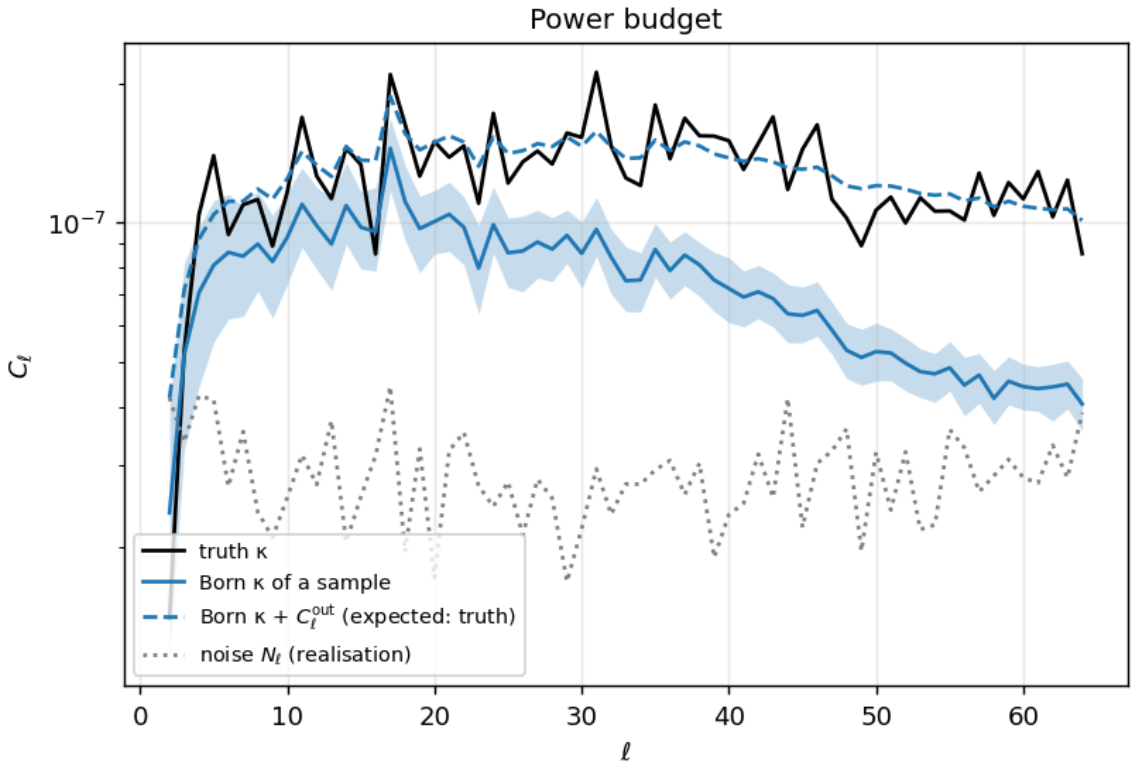
TEST

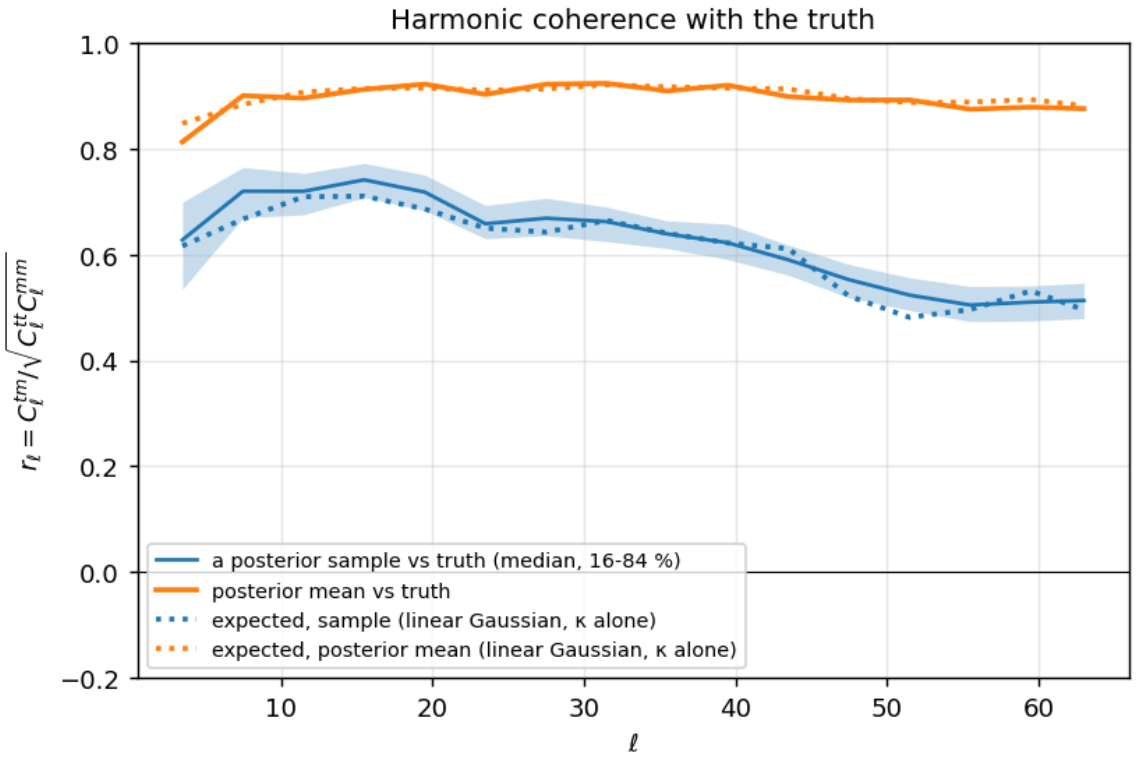
same κ likelihood: 300 noise draws on the noiseless map

ANSWER

no model error

κ alone reconstructs the map as a linear Gaussian model predicts





κ reconstructs the field in front of the galaxies

